

Course Syllabus

General Information

Faculty	Information Technology		
Department	Computer Science		
Semester	First	Academic Year	2025/2026

Course Details

Course Title	Robotics Programming			Course No.		0432405	
Type of Learning	☒ On- campus ☐ Blended (.....Platform) ☐ Online (.....Platform)						
Credit Hours	3	Theoretical	2	Practical	1		
Course Level (according to JNQF)			6	JNQF Hours*		12.3	
Pre-requisite	Python Programming- 0431201			Co-requisite		0432406	

Course Instructor Information

Instructor Name	Dr Duaa Mehiar		
Instructor E-mail	duaa.mehiar@meu.edu.jo		
Course Time			
Office Hours			
Office Number	N112	Office Phone	
Lab Instructor Name –If assigned-			

Sources and References

1. Textbook

Introduction to Robotics: Analysis, Control, Applications / Saeed B. Niku/ Wiley / 2019

2. References

- Learn Robotics Programming: Build and control autonomous robots using Raspberry Pi 3 and Python by Danny Staple, 2018.
- Modern Robotics: Mechanics, Planning, and Control 1st Edition by Kevin M. Lynch (Author), Frank C. Park (Author), 2017.
- Probabilistic Robotics (Intelligent Robotics and Autonomous Agents series) 1st Edition, 2005.
- Robot Dynamics and Control , Mark W. Spong and M. Vidyasagar, John Wiley & Sons.
- Robotics: Vision, and Control, Peter Corke, Springer,2011.

Teaching Methods

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|---------------------------|------------------------------|---------------------------|
| 1. Lecture-based learning | 2. Technology-based learning | 3. Group learning |
| 4. Individual learning | 5. Inquiry-based learning | 6. Expeditionary learning |

Course Description

Explore key concepts of artificial intelligence (AI) algorithms and programming for robot systems. Build a mental framework for AI in robotics; examine reactive sensing-acting loops; and design, implement, and evaluate autonomous systems and human-robot interaction. Consider machine learning and ethics; survey robotics fundamentals; and work with the basic components of robot systems. Practice robotics programming, computer vision, motion planning, mobile mechanisms, and manipulator kinematics. Engage with sensing, actuators, control theory, robot coordinate systems, differential motions, path control, dynamics, force control, and robot simulation. Investigate automation and robot programming languages; integrate actuators and sensors throughout.

Course Learning Outcomes (CLOs) and its correlation with Program Learning Outcomes (PLOs) and JNQF Descriptors

No.	CLOs	PLOs	JNQF Descriptors		
			Knowledge	Skill	Competency
1	Define the artificial intelligence robot for applications involving sensing, acting, planning, and learning, and Define human-robot interaction (HRI).	1	✓		
2	Analyze the Robotics Systems Architecture, components, and Kinematics.	1,2		✓	
3	Demonstrate the basic programming of a robotic system and Select sensors for performing robotic tasks.	1		✓	
4	Troubleshoot and resolve common problems related to the robotic system & motion planning problems.	2, 6			✓
5	Design and Implement a Robotics system to solve problems using various microcontroller Kits.	2, 6		✓	

Course Topics

Week	Topic	Lecture Type		CLOs	Assessment	
		Lecture No.	Lecture Type		Assessment Tool	Summative/ Formative
1	Course Syllabus Discussion & Introduction to the course Lab1: ROS 2 essentials & workspace	Lecture (1)	L	1	-	F
		Lecture (2)	L	1	-	F
2	Review Basic Robotics and AI Concepts Lab2: Core Architecture	Lecture (1)	L	1	1,3,4	S
		Lecture (2)	L	1	1,3,4	S
3	Legged and Wheeled robot's locomotion (Concepts & Examples) Lab3: Getting Started with ROS 2 on Linux	Lecture (1)	L	1	1,3	S
		Lecture (2)	L	2	1,3	S
	Mobile robot Kinematics	Lecture (1)	L	2	1,3	S

4	Robots Simulator Lab4: Developing ROS 2 Nodes:	Lecture (2)	P	3	1,3	S
5	Beyond Basic Kinematics, Motion Control (Kinematic Control)	Lecture (1)	L	2	1,3,4	S

	Robots Simulator Lab5: Service and Action Servers	Lecture (2)	P	3	1,3,4	S
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6	Perception, Sensors for Mobile Robots (Tactile, Finders, GPS) Software packages (Motion neurons sensors) Lab 6: Parameter Server	Lecture (1)	L	3	1,3,4	S
		Lecture (2)	P	5	1,3,4	S
7	Manipulator dynamics. Control systems for manipulators; Position control of manipulators; Control laws; Lab7: Create Custom ROS 2 Messages , Launch System (Files and RQT Tool)	Lecture (1)	L	3	1,3,4	
		Lecture (2)	P	4	1,3,4	S
8	Mid-Term	Lecture (1)	P	2	4	S
		Lecture (2)	L	-	-	S

Week	Topic	Lecture Type		CLOs	Assessment	
		Lecture No.	Lecture Type		Assessment Tool	Summative/ Formative
9	Practical considerations; Adaptive control of manipulators. Dynamic Analysis and Forces Lab 8: Visualization and Simulation	Lecture (1)	L	5	2,3,4	S
		Lecture (2)	P	5	2,3,4	S
10	Trajectory Planning 8 Motion trajectories: generation, path planning, and tracking Lab9: Rviz 2 ,Visualizing robot states	Lecture (1)	P	5	2,3,4	S
		Lecture (2)	P	5	2,3,4	S
11	Actuators Lab10: GazeboSim Harmonic	Lecture (1)	L	3	2, 4	S
		Lecture (2)	P	5	2, 4	S
12	Sensors, motors Perception. Robot's Sensors; Internal and External Sensors. Lab11: Importing 3D models into Gazebo	Lecture (1)	L	4	2, 4	S
		Lecture (2)	P	4	2, 4	S
13	Vision Systems Lab12 : project	Lecture (1)	L	4	2, 4	S
		Lecture (2)	P	4	2, 4	S
14	Robotics Applications: Chat-bots, Drones, VR, AR	Lecture (1)	L	2	2, 4	S
		Lecture (2)	P	4	2, 4	F
15	Project Presentation and Discussion Final Exam	Lecture (1)	P	1,2,3,4,5	4	F
		Lecture (2)	P	-		

Course Evaluation Time and Weight

Assessment Tool	Weight %	Expected Due Date
1 Mid Written Exam.	25	W8

2 Final Written Exam.	40	W16
3 Assignments & Quizzes	15	W3, W5, W7, W11
4 Project.	20	W14, W15

Course Policies

Item	Policy
Quizzes	No makeup quizzes.
	- A minimum of 2 quizzes are given. - Each Quiz is out of 5. - If 2 quizzes or more are given, then the lowest quiz's grade is dropped. - Pop quizzes may be administered periodically throughout the semester.
Exams	The format of the exams is generally (but NOT always) as follows: General Definitions, Multiple-Choice, True/False, Analyze a Problem, Essay Questions, etc.
Makeup Exams	Makeup exam should not be given unless there is a valid excuse.
Drop Date	Last day to drop the course is according to the University regulations.
Academic Honesty	- Cheating or copying on exam or quiz is an illegal and unethical activity. - Standard MEU policy will be applied. - All graded assignments must be your own work (your own words).
Attendance	- Excellent attendance is expected. - MEU policy requires the faculty member to assign ZERO grade if a student misses 10% of the classes. - Sign-in sheets will be circulated. - If you miss class, it is your responsibility to find out about any announcements or assignments you may have missed.
E-Learning	It is your responsibility to check the course's eLearning website regularly (at least once every 2-3 days) for announcements and material.
Workload	The student should expect to spend 6 hours per week as an average workload.
Graded Exams	Instructor should return exam papers graded to students within one week after the exam date.
Participation	- Participation in and contribution to class discussions will affect your final grade positively. Raise your hand if you have any question. - Making any kind of disruption and (side talks) in the class will affect you negatively.
Assignment/ Projects	- Homework's are as mentioned in the previous table. - Students organize themselves in teams of (2-3) students each. Each team will give at least one presentation regarding a selected case, delivering a fully working application. - More details about phases and topics of the projects will be given later on.
Others	

Rubrics and Feedbacks (Project)

Criteria	Excellent	Very Good	Adequate	Poor
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Creativity (Approach and Topic):	shows deep insight, and significant understanding of the problem; goes beyond the brief and demonstrates significant originality in the ideas and their application; uses precedents and/or materials in unexpected ways; surprising and delightful outcome. (5 marks)	shows engagement, exploration and insight; uses precedents and/or materials in relatively original ways. (4 marks)	satisfies minimum requirements of the brief and/or a minor increment over existing work (extension/adaptation of a precedent) (2 marks)	does not satisfy brief and no originality or creativity demonstrated (e.g. direct replication of prior approach without extension) (1 mark)
Context (Concept and Background):	provides a thoughtful and considered introduction to the work and supports the context with relevant references and critical analysis of past work.(6 marks)	provides a thoughtful and considered introduction to the work with limited references and limited analysis of past work. (4 marks)	satisfies minimum requirements. (2 marks)	does not satisfy brief (no mention of problem space, scenario) and/or does not include precedents. (1 marks)
Execution	provides a strong technical understanding; well considered component choices; functions and is well	functional; demonstrates understanding of components and applied them in a working	satisfies minimum requirements - provides the minimum core	does not satisfy brief - code & circuit not included or does
	organized; and works with advanced components, implements functionality beyond the brief and demonstrates technical skill , shows competence in the design and material choices (8 marks)	circuit; circuit is well organized; component selection is sensible and demonstrates some technical skill. that form is sensible and appropriate. (5 marks)	functionality, is basic in operation and could be improved. (3 marks)	not compile. (1 mark)
Communication (Documentation, Presentation):	as good, but includes an additional level of rigor, reflection or professionalism that elevates the outcome. e.g. Documentation that is ready to share online for press or crowdfunding. (6 marks)	clearly communicated and sufficiently detailed to easily repeat the outcome; well and appropriately illustrated with media, diagrams and code. (4 marks)	satisfies minimum requirements - it contains the required components; but reasonably poor quality, verbose, unclear or shows other communication issues. (2 marks)	documentation is missing or doesn't provide any illustration (1 mark)

*** JNQF Hours Calculation**

Activity	Duration/ week	Times/ Semester	No. of Hours
Learning Activities			
Lectures and Seminars	2	14	28
Tutorials	1	10	10
laboratory	2	13	26
Activities for Assessment and Evaluation			
First Exam/ Midterm	1	1	1
Second Exam (If applicable)	-	-	-
Final Exam	2	1	2
Quizzes	0.5	2	1

Self-Learning Activities			
Assignments/ Homework	3	2	6
Case Studies	2	2	4
Presentations	2	2	4
E-learning	1.5	14	21
Extra Readings	2	10	20
Total Hours	123		
JNQF Hours	=Total Hours/10 =12.3		